

Peter A. Lilley

Software Engineer - C#/.NET and C++17 Systems, Geometry and Automation Tooling

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SUMMARY

Software engineer. Writes C#, C++17, and Python against the Civil 3D, AutoCAD, and Revit APIs. Wrote C++ desktop applications and engine scripting before LLM tools existed. Constrains model output to a typed command schema, so deterministic code owns anything that has to be correct. Focus is applied ML systems and inference integration, not model training.

SKILLS

Languages: C#, Python, C++, Dart, TypeScript, JavaScript, SQL, LISP, VBA

Applied ML and inference: Structured model output with validation gates, Tesseract OCR, Levenshtein and fuzzy string matching, retrieval over structured corpora, COLMAP and instant-ngp photogrammetry, containerized inference, edge deployment, Ollama, LM Studio

Databases and Data Systems: SQLite, Firebase, Postgres, XData schemas, data modeling, JSON structuring, graph calculators

Engineering and math: Matrix math, 3D graphics, Express, React, Git, CMake, Linux, macOS, Docker, Bash scripting for data pipelines

Domain systems, quality, and networking: CAD plugins, Revit automation, 3D metadata, ARCore, part dimensioning, coordinate measuring machines, ISO 9001, TCP/IP, IPv4/IPv6 subnetting, Cisco routing and switching

EXPERIENCE

David Mason & Associates - Staff Technician (CAD automation / Civil 3D tools)

Jul 2024 - Present

St. Louis, MO

- Built a hydraulic grade line tool for Civil 3D, shipped as one DLL with a C++ core wrapped in C# and a WinForms interface. It draws pipe network water levels from an Excel source and edits pipe properties in bulk. Staff engineers documented the manual method at 4 to 8 hours per run and confirmed the tool does the same work in a few clicks.
- Built layout-wide edit commands that delete or move matching objects across every layout in a drawing, and freeze layers by drawing scale. A local phrase-to-command lookup picks the target layouts. Peers confirmed the manhour savings in a formal civil workload review.
- Write C# plugins against the Civil 3D .NET API to automate layout, audit drawings, and batch-rename title blocks. Typed metadata schemas attach to model entities and carry company standards across disciplines.
- Built OCR ingestion software that parses existing PDF plan sets, checks values, and populates drawings.
- Automate repetitive drafting with Python and Dynamo. In self-timed runs, one plan-production cycle dropped from 8 to 12 hours to about 30 seconds.
- Constrain model output to a typed command schema. Deterministic C# code then constructs and validates all geometry, so a bad request fails validation instead of shipping a plausible wrong number.

Component Bar Products - Quality Engineer - Manufacturing / Precision Machining

Apr 2023 - Jun 2024

O'Fallon, MO

- Programmed and operated coordinate measuring machines. Verified precision-machined parts against GD&T specifications.
- Validated dimensional tolerances against CAD models and prints to catch geometry mismatches.
- Enforced ISO 9001 across production lines. Managed calibration and traceability of metrology assets.

Menard, Inc. - Assistant Department Manager - Building Materials and Millwork

Jun 2020 - Mar 2023

Manchester / Ballwin, MO

- Priced and estimated large commercial material orders. Supervised department staff.

Heideman & Associates, Inc. - Revit/CAD Technician - Mechanical/Plumbing

May 2018 - Apr 2019

Fenton, MO

- Produced MEP construction documents. Built as-built models in Revit and AutoCAD.
- Built automated data libraries for building-system objects. Standardized outdated details.

PROJECTS

Digital Twin Pro

Personal product, Free Parameter LLC | Flutter/Dart 3, SQLite, Firebase, Google Gemini API, ARCore | github.com/freeParameterized/digital-twin-pro

- Designed and shipped a cross-platform 3D inventory application to a Google Play beta.
- Architected a custom 3D renderer with a Z-sorted draw queue and a custom projection instead of a game engine. Used LLM assistance to write most of the code.
- Added photo-based item detection through a paid Gemini API.
- Stored inventory in SQLite with a move audit log, QR booklets, and CSV/JSON export.

CAD integration bridge

C++17, Dear ImGui, Windows COM | [private repository](#), [walkthrough on request](#)

- Built a C++17 Dear ImGui host. It bridges BricsCAD, AutoCAD, and Civil 3D through Windows COM.
- It discovers installed SDKs at runtime and hot-loads LISP, C#, and BRX plugins.
- This personal project proved the architecture before a similar commercial deployment.

Interactive Portfolio (this site)

React, TypeScript, Three.js, Express, Ollama, Docker

- Built a React and TypeScript front end over an Express API.
- A local model answers questions from structured resume data. The site works statically when the model server is offline.
- Containerized the inference stack with Docker for Linux, macOS, and Windows hosts.

EDUCATION

Calculus II (CLEP Credit) - Saint Louis University (SLU)

2018

St. Louis, MO

Accepted by Ranken Technical College for their calculus curriculum.

Coursework toward Latin American Studies with Technical Applications - Missouri University of Science and Technology

2019 - 2020

Rolla, MO

Coursework included no-calculator College Algebra. It covered determinants, Cramer's rule, and introductory linear algebra concepts. Technical fluency in Spanish.

A.A.S., Building Systems Engineering Technology - Ranken Technical College

2019

Wentzville, MO